

AWS Practice Questions

PRACTICE QUESTIONS (15 Scenario-Based Exercises)

Each scenario includes an answer/guidance section.

1. Launching a Highly Available Web Server on EC2

Scenario:

Your company wants to deploy a web application running on Apache. It must remain available even if one Availability Zone fails. Configure the required EC2 infrastructure.

Answer:

Create an Auto Scaling Group (ASG) across two or more AZs with a Launch Template installing Apache via user data. Attach an Application Load Balancer (ALB) to distribute traffic.

2. Secure Access to EC2 Instances

Scenario:

Developers require SSH access to EC2 instances only from your office IP. Configure the security properly.

Answer:

Edit the EC2 Security Group inbound rule to allow port 22 only from the office public IP. Avoid using 0.0.0.0/0.

3. Reduce Cost for a Background Compute Job

Scenario:

A nightly batch job runs for 2 hours on EC2. Recommend a cost-effective instance option.

Answer:

Use EC2 Spot Instances or Scheduled Reserved Instances since the workload is interruptible and predictable.

4. Migrate a Single-AZ RDS to Multi-AZ

Scenario:

Your RDS MySQL instance currently runs in Single-AZ mode. Management wants automatic failover.

Answer:

Modify the RDS instance and enable Multi-AZ Deployment, which creates a synchronous standby.

5. Improve RDS Read Scalability

Scenario:

A reporting team sends heavy read queries and slows down the main DB. Suggest a solution.

Answer:

Add RDS Read Replicas for scaling read workloads.

6. Store Database Password Securely for EC2 Applications

Scenario:

An EC2-based application needs credentials to connect to RDS. Storing them in code is not allowed.

Answer:

Use AWS Secrets Manager or SSM Parameter Store and grant access using IAM roles.

7. Deploy a Simple Java Application on Elastic Beanstalk

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Scenario:

You must deploy a Java app with minimal provisioning overhead.

Answer:

Create an Elastic Beanstalk Java environment, upload the application ZIP, and EB handles provisioning of EC2 + ALB + Auto Scaling.

8. Version Control for Application Deployments in Elastic Beanstalk

Scenario:

You want to roll back to an older application version quickly.

Answer:

Elastic Beanstalk keeps Application Versions in S3. Choose a previous version and redeploy.

9. Containerizing an Application for ECS

Scenario:

You need to deploy a Node.js service using containers in ECS. Start by preparing the image.

Answer:

Create a Dockerfile, build the image, push it to Amazon ECR, then reference it in your ECS Task Definition.

10. Deploying ECS Service with High Availability

Scenario:

Run your ECS service so it remains available even during instance failures.

Answer:

Use ECS with Fargate or EC2 launch type in a cluster spanning multiple AZs with a service + load balancer.

11. Continuous Deployment to ECS Using ECR

Scenario:

Developers push new Docker images frequently. Automate ECS deployment upon new image push.

Answer:

Use Amazon ECR Image Push → EventBridge → CodePipeline/CodeDeploy → ECS Service Update.

12. Reduce RDS Costs Without Downtime

Scenario:

RDS instance is underutilized, but production must stay online. How to reduce cost?

Answer:

Modify instance type to a smaller class or switch to Serverless (Aurora) if supported.

13. EC2 Instance Termination Protection

Scenario:

A critical instance was accidentally terminated. Prevent this going forward.

Answer:

Enable Termination Protection on the EC2 instance and enforce IAM permissions.

14. Automatic EC2 Recovery for Hardware Failure

Scenario:

Your EC2 instance must auto-recover if the underlying hardware has issues.

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Answer:

Enable EC2 Auto Recovery using CloudWatch alarms.

15. Blue/Green Deployment for ECS Application

Scenario:

You need zero-downtime deployments for an ECS app.

Answer:

Use ECS + CodeDeploy Blue/Green Strategy with ALB routing.

MULTIPLE-CHOICE QUESTIONS (15 MCQs with Answers & Explanations)

1. Which service provides resizable compute capacity in the cloud?

- A. Amazon RDS
- B. Amazon EC2
- C. AWS Lambda
- D. Amazon S3

Correct Answer: B

Explanation: EC2 provides virtual machines with resizable compute capacity.

2. Which feature distributes incoming traffic across EC2 instances?

- A. Auto Scaling
- B. IAM
- C. Elastic Load Balancing
- D. SQS

Correct Answer: C

Explanation: ELB balances traffic across multiple EC2 instances.

3. How can you securely store database credentials used by an EC2 instance?

- A. Store in user data
- B. Store in a text file on EC2
- C. Use Secrets Manager
- D. Embed in source code

Correct Answer: C

Explanation: Secrets Manager securely stores and rotates secrets.

4. What does Multi-AZ provide for RDS?

- A. Better read performance
- B. Automatic failover
- C. Cheaper prices
- D. Faster queries

Correct Answer: B

Explanation: Multi-AZ creates a standby instance for high availability.

5. Which RDS feature improves read performance?

- A. Multi-AZ
- B. Read Replicas

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- C. DB Snapshots
- D. IAM roles

Correct Answer: B

Explanation: Read Replicas handle read-heavy workloads.

6. Which service automatically provisions EC2, ELB, scaling, and deployment?

- A. ECS
- B. Elastic Beanstalk
- C. CloudFormation
- D. Lambda

Correct Answer: B

Explanation: Elastic Beanstalk automates application provisioning.

7. Where are Elastic Beanstalk application versions stored?

- A. Amazon S3
- B. Amazon EBS
- C. AWS Artifact
- D. Lambda Layers

Correct Answer: A

Explanation: EB stores versions in an S3 bucket.

8. Amazon ECR is best described as:

- A. A compute engine for running containers
- B. A container registry
- C. A database service
- D. A load balancer

Correct Answer: B

Explanation: ECR is a fully managed Docker container registry.

9. ECS Tasks are defined using:

- A. IAM Policies
- B. CloudTrail Logs
- C. Task Definitions
- D. Route Tables

Correct Answer: C

Explanation: Task Definitions describe container configuration.

10. Which ECS launch type removes the need to manage EC2 instances?

- A. EC2
- B. Spot
- C. Fargate
- D. On-Demand

Correct Answer: C

Explanation: Fargate is serverless for containers.

11. Which EC2 pricing option is most cost-effective for flexible workloads?

- A. On-Demand
- B. Reserved

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- C. Dedicated Hosts
- D. Spot Instances

Correct Answer: D

Explanation: Spot Instances offer up to 90% discounts for interruptible workloads.

12. Which Security Group rule allows HTTP traffic?

- A. Port 22
- B. Port 443
- C. Port 80
- D. Port 3389

Correct Answer: C

Explanation: HTTP operates on port 80.

13. What does an ECS Service ensure?

- A. Code compilation
- B. Desired number of tasks always running
- C. S3 bucket replication
- D. Database backup scheduling

Correct Answer: B

Explanation: ECS service ensures tasks maintain desired count.

14. Where are Docker images stored for ECS deployments?

- A. CloudTrail
- B. EC2
- C. S3 only
- D. ECR

Correct Answer: D

Explanation: ECR stores container images.

15. What is required to push images to ECR?

- A. EC2 Metadata
- B. IAM permissions
- C. Route 53
- D. RDS Proxy

Correct Answer: B

Explanation: Pushing images requires authenticated IAM permissions.